

# Problem Set 8

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## Problem 4: Data Generation

I generated a dataset with  $N = 100,000$  observations and  $K = 10$  variables. The matrix  $X$  has a column of ones as its first column and normally distributed random numbers in the remaining columns. The error vector  $\varepsilon \sim N(0, 0.25)$ , and  $Y = X\beta + \varepsilon$  where the true  $\beta$  is:

$$\beta = [1.5 \quad -1 \quad -0.25 \quad 0.75 \quad 3.5 \quad -2 \quad 0.5 \quad 1 \quad 1.25 \quad 2]'$$

## Problem 5: Closed-Form OLS

I computed  $\hat{\beta}_{OLS} = (X'X)^{-1}X'Y$  directly. The estimates are extremely close to the true  $\beta$ , which is expected given the large sample size of  $N = 100,000$ .

## Problem 6: Gradient Descent OLS

I computed  $\hat{\beta}_{OLS}$  using gradient descent with a learning rate of 0.0000003 and 10,000 iterations. The estimates are very close to the true  $\beta$ , though slightly less precise than the closed-form solution due to the finite number of iterations.

## Problem 7: nloptr L-BFGS and Nelder-Mead OLS

I computed  $\hat{\beta}_{OLS}$  using both the L-BFGS and Nelder-Mead algorithms in `nloptr`. With a sufficient number of iterations (`maxeval = 1,000,000`), both algorithms converge to the same solution, which closely matches the true  $\beta$ . Without enough iterations, Nelder-Mead performs worse than L-BFGS because it is a derivative-free method that converges more slowly in high dimensions.

## Problem 8: MLE with L-BFGS

I computed  $\hat{\beta}_{MLE}$  using the L-BFGS algorithm with the provided gradient function. The beta estimates are nearly identical to those from OLS, and the estimated  $\hat{\sigma} \approx 0.501$ , which is very close to the true value of  $\sigma = 0.5$ . A lower bound of 0.001 was imposed on  $\sigma$  to ensure the optimizer stayed in a valid region.

## Problem 9: OLS via `lm()`

Table 1 reports the OLS estimates from `lm(Y ~ X - 1)`. All ten estimates are extremely close to the true  $\beta$  values, with differences of at most 0.003 in absolute value. This is expected: with  $N = 100,000$  observations, the OLS estimator is very precise and the sampling variance is negligible. The  $R^2$  of 0.991 confirms that the model fits the data well, and the RMSE of 0.50 closely matches the true  $\sigma = 0.5$ .

|          | (1)               |
|----------|-------------------|
| X1       | 1.502<br>(0.002)  |
| X2       | -1.000<br>(0.002) |
| X3       | -0.251<br>(0.002) |
| X4       | 0.752<br>(0.002)  |
| X5       | 3.500<br>(0.002)  |
| X6       | -1.999<br>(0.002) |
| X7       | 0.499<br>(0.002)  |
| X8       | 1.001<br>(0.002)  |
| X9       | 1.247<br>(0.002)  |
| X10      | 2.002<br>(0.002)  |
| Num.Obs. | 100 000           |
| R2       | 0.991             |
| R2 Adj.  | 0.991             |
| AIC      | 145 726.7         |
| BIC      | 145 831.4         |
| Log.Lik. | -72 852.364       |
| RMSE     | 0.50              |